Linear Algebra and Its Applications

Section 6.5: Least-Squares Problems

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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6.5 Least-squares problems

A least-squares solution $\widehat{x}$ minimizes the *output error* $\|b - Ax\|$, not $\|b - x\|$. The minimizing output $\widehat{b} = A\widehat{x}$ is the orthogonal projection of b onto $\operatorname{Col} A$. The residual $r = b - A\widehat{x}$ is orthogonal to every column, so

$$A^T r = 0 \quad \Longleftrightarrow \quad A^T A\widehat{x} = A^T b.$$

These are the normal equations. They always have solutions, but the coefficient vector need not be unique when A has dependent columns. The projection is still unique. With full column rank and a QR factorization $A = QR$, solve $R\widehat{x} = Q^T b$ by back substitution; do not explicitly form an inverse for numerical work.

Practice problem 6.5.P1: A least-squares problem can have zero error

For the given matrix, direct multiplication gives

$$A(2, -1, 0)^T = (5, -3, -5)^T = b.$$

Thus $\widehat{x} = (2, -1, 0)^T$ is already an exact solution and therefore a least-squares solution: its residual is zero, and no norm can be smaller than zero. Its associated least-squares error is 0. For a normal-equation check,

$$A^T A = \begin{bmatrix} 3 & 9 & 0 \\ 9 & 83 & 28 \\ 0 & 28 & 14 \end{bmatrix}, \quad A^T b = (-3, -65, -28)^T,$$

and the chosen vector satisfies that system as well.

Practice problem 6.5.P2: An orthogonal target has zero as its projection

If b is orthogonal to all columns, $A^T b = 0$. The projection of b onto $\operatorname{Col} A$ is zero, so $x = 0$ is a least-squares solution and the minimum error is $\|b\|$. More precisely all least-squares solutions satisfy $Ax = 0$; thus they form $\operatorname{Nul} A$. The coefficient vector is unique only when that null space is $\{0\}$.

Exercise 6.5.1: Construct and solve the normal equations

(a) Compute the two sides from the original data.

$$A = \begin{bmatrix} -1 & 2 \\ 2 & -3 \\ -1 & 3 \end{bmatrix}, \qquad b = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}.$$

Taking column dot products gives

$$A^T A = \begin{bmatrix} 6 & -11 \\ -11 & 22 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} -4 \\ 11 \end{bmatrix}.$$

The normal equation is $A^T A\widehat{x} = A^T b$ with these entries.

(b) Solve that smaller system.

$$[A^TA \mid A^Tb] \sim \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \end{bmatrix}, \qquad \widehat{x} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

There is a pivot in both unknown columns, so the least-squares coefficient vector is unique. Its residual is

$$r = b - A\widehat{x} = \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix}.$$

Substituting into A^Tr gives zero, verifying the orthogonality condition independently. This certifies that the solution minimizes the error, even when $A\widehat{x}$ does not equal b.

Exercise 6.5.2: Construct and solve the normal equations

(a) Compute the two sides from the original data.

$$A = \begin{bmatrix} 2 & 1 \\ -2 & 0 \\ 2 & 3 \end{bmatrix}, \qquad b = \begin{bmatrix} -5 \\ 8 \\ 1 \end{bmatrix}.$$

Taking column dot products gives

$$A^TA = \begin{bmatrix} 12 & 8 \\ 8 & 10 \end{bmatrix}, \qquad A^Tb = \begin{bmatrix} -24 \\ -2 \end{bmatrix}.$$

The normal equation is $A^TA\widehat{x} = A^Tb$ with these entries.

(b) Solve that smaller system.

$$[A^TA \mid A^Tb] \sim \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & 3 \end{bmatrix}, \qquad \widehat{x} = \begin{bmatrix} -4 \\ 3 \end{bmatrix}.$$

There is a pivot in both unknown columns, so the least-squares coefficient vector is unique. Its residual is

$$r = b - A\widehat{x} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Substituting into A^Tr gives zero, verifying the orthogonality condition independently. This certifies that the solution minimizes the error, even when $A\widehat{x}$ does not equal b.

Exercise 6.5.3: Construct and solve the normal equations

(a) Compute the two sides from the original data.

$$A = \begin{bmatrix} 1 & -2 \\ -1 & 2 \\ 0 & 3 \\ 2 & 5 \end{bmatrix}, \qquad b = \begin{bmatrix} 3 \\ 1 \\ -4 \\ 2 \end{bmatrix}.$$

Taking column dot products gives

$$A^T A = \begin{bmatrix} 6 & 6 \\ 6 & 42 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 6 \\ -6 \end{bmatrix}.$$

The normal equation is $A^T A\widehat{x} = A^T b$ with these entries.

(b) Solve that smaller system.

$$[A^T A \mid A^T b] \sim \begin{bmatrix} 1 & 0 & \frac{4}{3} \\ 0 & 1 & -\frac{1}{3} \end{bmatrix}, \qquad \widehat{x} = \begin{bmatrix} \frac{4}{3} \\ -\frac{1}{3} \end{bmatrix}.$$

There is a pivot in both unknown columns, so the least-squares coefficient vector is unique. Its residual is

$$r = b - A\widehat{x} = \begin{bmatrix} 1 \\ 3 \\ -3 \\ 1 \end{bmatrix}.$$

Substituting into $A^T r$ gives zero, verifying the orthogonality condition independently. This certifies that the solution minimizes the error, even when $A\widehat{x}$ does not equal b.

Exercise 6.5.4: Construct and solve the normal equations

(a) Compute the two sides from the original data.

$$A = \begin{bmatrix} 1 & 3 \\ 1 & -1 \\ 1 & 1 \end{bmatrix}, \qquad b = \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}.$$

Taking column dot products gives

$$A^T A = \begin{bmatrix} 3 & 3 \\ 3 & 11 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 6 \\ 14 \end{bmatrix}.$$

The normal equation is $A^T A\widehat{x} = A^T b$ with these entries.

(b) Solve that smaller system.

$$[A^T A \mid A^T b] \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \qquad \widehat{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

There is a pivot in both unknown columns, so the least-squares coefficient vector is unique. Its residual is

$$r = b - A\widehat{x} = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}.$$

Substituting into $A^T r$ gives zero, verifying the orthogonality condition independently. This certifies that the solution minimizes the error, even when $A\widehat{x}$ does not equal b.

Exercise 6.5.5: Find every minimizing coefficient vector

The original data give

$$A^T A = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 2 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 14 \\ 4 \\ 10 \end{bmatrix}.$$

The reduced augmented normal matrix is

$$\begin{bmatrix} 1 & 0 & 1 & 5 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Set the free variable $x_3 = t$. Reading the first two equations gives

$$\widehat{x} = \begin{bmatrix} 5 - t \\ t - 3 \\ t \end{bmatrix}, \qquad t \in \mathbb{R}.$$

All these coefficient vectors produce the same fitted output $A\widehat{x} = \begin{bmatrix} 2 \\ 2 \\ 5 \\ 5 \end{bmatrix}$. The residual $\begin{bmatrix} -1 \\ 1 \\ 3 \\ -3 \end{bmatrix}$ is orthogonal to all columns of A. Conversely every minimizer solves the normal equations, so this family contains all least-squares solutions, not just selected ones.

Exercise 6.5.6: Find every minimizing coefficient vector

The original data give

$$A^T A = \begin{bmatrix} 6 & 3 & 3 \\ 3 & 3 & 0 \\ 3 & 0 & 3 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 27 \\ 12 \\ 15 \end{bmatrix}.$$

The reduced augmented normal matrix is

$$\begin{bmatrix} 1 & 0 & 1 & 5 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Set the free variable $x_3 = t$. Reading the first two equations gives

$$\widehat{x} = \begin{bmatrix} 5 - t \\ t - 1 \\ t \end{bmatrix}, \qquad t \in \mathbb{R}.$$

All these coefficient vectors produce the same fitted output $A\widehat{x} = \begin{bmatrix} 4 \\ 4 \\ 4 \\ 5 \\ 5 \\ 5 \end{bmatrix}$. The residual $\begin{bmatrix} 3 \\ -2 \\ -1 \\ 1 \\ 0 \\ -1 \end{bmatrix}$ is orthogonal to all columns of A. Conversely every minimizer solves the normal equations, so this family contains all least-squares solutions, not just selected ones.

Exercise 6.5.7: Compute the minimum error as a residual norm

Using the least-squares solution in Exercise 3, the residual is $r = \begin{bmatrix} 1 \\ 3 \\ -3 \\ 1 \end{bmatrix}$. Therefore

$$\|b - A\widehat{x}\| = \sqrt{(1)^2 + (3)^2 + (-3)^2 + (1)^2} = \boxed{2\sqrt{5}}.$$

The error is the norm, rather than its square. The sum of squared residuals is 20.

Exercise 6.5.8: Compute the minimum error as a residual norm

Using the least-squares solution in Exercise 4, the residual is $r = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$. Therefore

$$\|b - A\widehat{x}\| = \sqrt{(1)^2 + (1)^2 + (-2)^2} = \boxed{\sqrt{6}}.$$

The error is the norm, rather than its square. The sum of squared residuals is 6.

Exercise 6.5.9: Find a projection and its coefficient vector

The columns happen to be orthogonal, as shown by the diagonal Gram matrix:

$$A^T A = \begin{bmatrix} 14 & 0 \\ 0 & 42 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 4 \\ 6 \end{bmatrix}.$$

(a) Form the orthogonal projection.

For an orthogonal column list, $\widehat{b} = \sum_j (a_j^T b)/(a_j^T a_j)\, a_j$. Here those coefficients give

$$\widehat{b} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

(b) Record a least-squares input.

The same coefficients solve the diagonal normal equations:

$$\widehat{x} = \begin{bmatrix} \frac{2}{7} \\ \frac{1}{7} \end{bmatrix}.$$

Indeed $A\widehat{x} = \widehat{b}$, and the residual $b - \widehat{b} = \begin{bmatrix} 3 \\ -3 \\ -3 \end{bmatrix}$ has zero dot product with every column. The nonzero diagonal entries also make this input unique.

Exercise 6.5.10: Find a projection and its coefficient vector

The columns happen to be orthogonal, as shown by the diagonal Gram matrix:

$$A^T A = \begin{bmatrix} 3 & 0 \\ 0 & 24 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 9 \\ 12 \end{bmatrix}.$$

(a) Form the orthogonal projection.

For an orthogonal column list, $\widehat{b} = \sum_j (a_j^T b)/(a_j^T a_j)\, a_j$. Here those coefficients give

$$\widehat{b} = \begin{bmatrix} 4 \\ -1 \\ 4 \end{bmatrix}.$$

(b) Record a least-squares input.

The same coefficients solve the diagonal normal equations:

$$\widehat{x} = \begin{bmatrix} 3 \\ \frac{1}{2} \end{bmatrix}.$$

Indeed $A\widehat{x} = \widehat{b}$, and the residual $b - \widehat{b} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ has zero dot product with every column. The nonzero diagonal entries also make this input unique.

Exercise 6.5.11: Find a projection and its coefficient vector

The columns happen to be orthogonal, as shown by the diagonal Gram matrix:

$$A^TA = \begin{bmatrix} 54 & 0 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & 27 \end{bmatrix}, \qquad A^Tb = \begin{bmatrix} 36 \\ 0 \\ 9 \end{bmatrix}.$$

(a) Form the orthogonal projection.

For an orthogonal column list, $\widehat{b} = \sum_j (a_j^Tb)/(a_j^Ta_j)\, a_j$. Here those coefficients give

$$\widehat{b} = \begin{bmatrix} 3 \\ 1 \\ 4 \\ -1 \end{bmatrix}.$$

(b) Record a least-squares input.

The same coefficients solve the diagonal normal equations:

$$\widehat{x} = \begin{bmatrix} \frac{2}{3} \\ 0 \\ \frac{1}{3} \end{bmatrix}.$$

Indeed $A\widehat{x} = \widehat{b}$, and the residual $b - \widehat{b} = \begin{bmatrix} 6 \\ -1 \\ -4 \\ 1 \end{bmatrix}$ has zero dot product with every column. The nonzero diagonal entries also make this input unique.

Exercise 6.5.12: Find a projection and its coefficient vector

The columns happen to be orthogonal, as shown by the diagonal Gram matrix:

$$A^TA = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}, \qquad A^Tb = \begin{bmatrix} 1 \\ 14 \\ -5 \end{bmatrix}.$$

(a) Form the orthogonal projection.

For an orthogonal column list, $\widehat{b} = \sum_j (a_j^Tb)/(a_j^Ta_j)\, a_j$. Here those coefficients give

$$\widehat{b} = \begin{bmatrix} 5 \\ 2 \\ 3 \\ 6 \end{bmatrix}.$$

(b) Record a least-squares input.

The same coefficients solve the diagonal normal equations:

$$\widehat{x} = \begin{bmatrix} \frac{1}{3} \\ \frac{14}{3} \\ -\frac{5}{3} \end{bmatrix}.$$

Indeed $A\widehat{x} = \widehat{b}$, and the residual $b - \widehat{b} = \begin{bmatrix} -3 \\ 3 \\ 3 \\ 0 \end{bmatrix}$ has zero dot product with every column. The nonzero diagonal entries also make this input unique.

Exercise 6.5.13: Disqualify one candidate by finding a better one

Direct multiplication gives $Au = (11, -11, 11)^T$ and $Av = (7, -12, 7)^T$. With $b = (11, -9, 5)^T$, the residuals are $(0, 2, -6)^T$ and $(4, 3, -2)^T$, respectively. Their squared norms are 40 and 29. Since the candidate v has strictly smaller error than u, the vector u cannot minimize the error. This proves u is not a least-squares solution without solving the least-squares problem. It does not assert that v is a minimizer.

Exercise 6.5.14: Equal candidate errors do not prove minimality

Multiplication gives $Au = (3, 8, 2)^T$, $Av = (7, 2, 8)^T$. The two residuals from $b = (5, 4, 4)^T$ are $(2, -4, 2)^T$ and $(-2, 2, -4)^T$, both with squared norm 24. Now take the midpoint $z = (u + v)/2 = (5, -5)^T$. Linearity gives $Az = (5, 5, 5)^T$, whose residual is $(0, -1, -1)^T$, with squared norm $2 < 24$. Thus *neither* u nor v can be a least-squares solution. Exhibiting this one better candidate suffices; we have not computed the actual minimizer.

Exercise 6.5.15: Use the supplied QR factorization

The columns of Q are orthonormal, so solve $R\widehat{x} = Q^T b$. Here

$$Q^T b = \begin{bmatrix} (14 + 6 + 1)/3 \\ (-7 + 6 - 2)/3 \end{bmatrix} = \begin{bmatrix} 7 \\ -1 \end{bmatrix}, \qquad R = \begin{bmatrix} 3 & 5 \\ 0 & 1 \end{bmatrix}.$$

Back substitution first gives $\widehat{x}_2 = -1$, then $3\widehat{x}_1 + 5(-1) = 7$, so $\widehat{x}_1 = 4$.

Conclusion. $\widehat{x} = (4, -1)^T$. No explicit inverse of R is needed.

Exercise 6.5.16: Back substitute in another triangular system

The given columns of Q give

$$Q^T b = \begin{bmatrix} (-1 + 6 + 5 + 7)/2 \\ (1 + 6 - 5 + 7)/2 \end{bmatrix} = \begin{bmatrix} 17/2 \\ 9/2 \end{bmatrix}.$$

Thus $5\widehat{x}_2 = 9/2$ and $2\widehat{x}_1 + 3\widehat{x}_2 = 17/2$. These yield $\widehat{x}_2 = 9/10$, $\widehat{x}_1 = 29/10$.

Conclusion. $\widehat{x} = (2.9, 0.9)^T$, the unique least-squares solution.

Exercise 6.5.17: Five least-squares assertions

(a) **True.** This is the defining minimization problem in the Euclidean output norm.

(b) **True.** The nearest output in $\operatorname{Col} A$ is precisely $\widehat{b} = \operatorname{proj}_{\operatorname{Col} A} b$.

(c) **False.** The displayed inequality is reversed. A minimizer satisfies $\|b - A\widehat{x}\| \leq \|b - Ax\|$ for every x, not the other way around.

(d) **True.** The normal equations are equivalent to $A^T(b - A\widehat{x}) = 0$, the orthogonal-projection condition.

(e) **True.** Independent columns make A^TA invertible, so the always-consistent normal equations have exactly one solution.

Exercise 6.5.18: Distinguish input coefficients from fitted outputs

(a) **True.** An exact solution has zero error, the smallest possible.

(b) **False.** The point in $\operatorname{Col} A$ closest to b is $A\widehat{x}$, not generally $\widehat{x}$, which has n input coordinates rather than m output coordinates.

(c) **True.** The entries of $\widehat{x}$ are the weights whose column combination produces the projected output.

(d) **False without a rank assumption.** A^TA may be singular, so the displayed inverse may not exist. For the zero matrix every input is a minimizer, but A^TA is not invertible in positive dimension.

(e) **False as a numerical claim.** In exact arithmetic they characterize minimizers, but forming A^TA can magnify conditioning problems and roundoff. QR methods are often more reliable.

(f) **False as stated about the best computation.** With full column rank, the formula is algebraically valid, but solving $R\widehat{x} = Q^Tb$ by back substitution is preferable to explicitly forming R^{-1}. If full column rank is absent, the stated inverse may not exist at all.

Exercise 6.5.19: Prove equality of two null spaces

(a) Start with $Ax = 0$.

Multiplying by A^T gives $A^TAx = A^T0 = 0$.

(b) Start with $A^TAx = 0$.

Multiplying on the left by x^T gives $0 = x^TA^TAx = (Ax)^T(Ax) = \|Ax\|^2$. A sum of squares is zero only when every entry is zero, so $Ax = 0$.

Conclusion. Both inclusions hold, hence $\operatorname{Nul} A = \operatorname{Nul}(A^TA)$.

Exercise 6.5.20: Invertibility of the Gram matrix forces independent columns

If $Ax = 0$, Exercise 19 gives $A^TAx = 0$. Invertibility of A^TA forces $x = 0$, so the columns of A are independent. There is no step requiring A itself to be square or invertible.

Exercise 6.5.21: The converse and its rank consequences

(a) Independent columns give $\operatorname{Nul} A = \{0\}$. Exercise 19 makes $\operatorname{Nul}(A^TA) = \{0\}$ too. Since A^TA is square $n \times n$, the Invertible Matrix Theorem makes it invertible.

(b) There are n independent columns in $\mathbb{R}^m$, so $n \leq m$. Equivalently, n pivot columns require at least n rows.

(c) All n columns are pivot columns, so $\operatorname{rank} A = n$.

Exercise 6.5.22: The Gram matrix has the same rank

Both A and A^TA have n columns, so rank-nullity gives $\operatorname{rank} A = n - \dim \operatorname{Nul} A$ and $\operatorname{rank}(A^TA) = n - \dim \operatorname{Nul}(A^TA)$. The null spaces are equal by Exercise 19, so the subtracted dimensions are equal. Thus $\operatorname{rank}(A^TA) = \operatorname{rank} A$, even when the columns are dependent.

Exercise 6.5.23: Projection without an orthogonal column basis

Independent columns make A^TA invertible. Solving the normal equations gives $\widehat{x} = (A^TA)^{-1}A^Tb$. Multiply by A to get the fitted output:

$$\boxed{\widehat{b} = A(A^TA)^{-1}A^Tb}.$$

This is a formula for the projection in $\mathbb{R}^m$, not just its coefficient vector in $\mathbb{R}^n$. Orthogonality follows by computing $A^T(b - \widehat{b}) = A^Tb - A^TA(A^TA)^{-1}A^Tb = 0$.

Exercise 6.5.24: Orthonormal columns simplify the normal equations

If the columns are orthonormal, their Gram matrix is $A^TA = I_n$. Thus the normal equations reduce immediately to $\boxed{\widehat{x} = A^Tb}$. The corresponding projection is AA^Tb. Full column rank is automatic because no nontrivial combination of orthonormal columns can be zero.

Exercise 6.5.25: Fit two incompatible equations with a common left side

Let $s = x + y$. The squared error is

$$(2 - s)^2 + (4 - s)^2 = 2(s - 3)^2 + 2.$$

This is minimized exactly when $s = 3$. Hence all least-squares solutions are $(x, y) = (t, 3 - t)$, $t \in \mathbb{R}$. They all produce the projected output $(3, 3)^T$ and error $\sqrt{2}$. The fitted output is unique, while the two input weights are not.

Exercise 6.5.26: Design the approximate low-pass filter from the actual table

Use the table's rounded value $q = 0.7$, and define the periodic lists

$$y = (1, q, 0, -q, -1, -q, 0, q), \qquad w = (1, -q, 0, q, -1, q, 0, -q).$$

Indices are reduced modulo 8. Form an eight-row block $Y_k = (y_{k+2}, y_{k+1}, y_k)$ and another block $Z_k = (w_{k+2}, w_{k+1}, w_k)$ for $k = 0, \dots, 7$. Then the requested 16×3 system is explicitly

$$A = \begin{bmatrix} Y \\ Z \end{bmatrix}, \quad x = (a_0, a_1, a_2)^T, \quad b = (q, 0, -q, -1, -q, 0, q, 1, 0, 0, 0, 0, 0, 0, 0, 0)^T.$$

For clarity the two blocks are

$$Y = \begin{bmatrix} 0 & q & 1 \\ -q & 0 & q \\ -1 & -q & 0 \\ -q & -1 & -q \\ 0 & -q & -1 \\ q & 0 & -q \\ 1 & q & 0 \\ q & 1 & q \end{bmatrix}, \quad Z = \begin{bmatrix} 0 & -q & 1 \\ q & 0 & -q \\ -1 & q & 0 \\ q & -1 & q \\ 0 & q & -1 \\ -q & 0 & q \\ 1 & -q & 0 \\ -q & 1 & -q \end{bmatrix}.$$

Column dot products give $A^T A = (198/25) I_3$ and $A^T b = (14/5, 99/25, 14/5)^T$. The diagonal normal equations therefore yield

$$\boxed{a_0 = a_2 = 35/99 \approx 0.35353535, \qquad a_1 = 1/2}.$$

The normal-equation residual is exactly zero with these coefficients and the rational table entry $q = 7/10$. More generally the same computation for a table value q gives $a_0 = a_2 = q/(1 + 2q^2)$ and $a_1 = 1/2$. Putting $q = \sqrt{2}/2$ gives $a_0 = a_2 = \sqrt{2}/4$; using $q = 0.707$ gives a very close approximation. The rounded-table answer should not be silently replaced by the exact-trigonometric answer.